AMATH/PMATH 331: Applied Real Analysis Fall 2026

Assignment 3a Submission; due Tuesday, October 20, 2026

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[1] Let $f : A \subseteq \mathbb{R}^n \to \mathbb{R}^m$.

[a] Let $B \subseteq \mathbb{R}^m$. Show that $f^{-1}(B^c) = A \cap (f^{-1}(B))^c$.

Answer:

Proof. Take arbitrary $x \in f^{-1}(B^c)$.

$$\begin{aligned}
&\iff x \in A \wedge f(x) \in B^c \\
&\iff x \in A \wedge f(x) \notin B \\
&\iff x \in A \wedge x \notin f^{-1}(B) \\
&\iff x \in A \wedge x \in (f^{-1}(B))^c \\
&\iff x \in A \cap (f^{-1}(B))^c
\end{aligned}$$

□

[b] Prove that f is continuous on A if and only if:

$$\begin{gathered}
\text{for every } \textit{closed} \text{ subset } F \subseteq \mathbb{R}^m, \text{ then} \\
f^{-1}(F) = A \cap E \text{ for some } \textit{closed} \text{ subset } E \subseteq \mathbb{R}^n.
\end{gathered} \tag{1}$$

(Note that if the domain A is all of $\mathbb{R}^n$, we can simplify this to $f^{-1}(F)$ is closed in $\mathbb{R}^n$ for every F closed in $\mathbb{R}^m$.)

Answer:

Proof. Suppose (1) holds. We have to show f is continuous on A.
Let arbitrary open $V \subseteq \mathbb{R}^m$. Then V^c is closed.
By (1), $f^{-1}(V^c) = A \cap E$ for some closed $E \subseteq \mathbb{R}^n$.
By [a], $f^{-1}(V^c) = A \cap (f^{-1}(V))^c$

$$\begin{aligned}
&A \cap E = A \cap (f^{-1}(V))^c \\
&A \setminus E = A \setminus (f^{-1}(V))^c \qquad\qquad f^{-1}(V) \subseteq A \\
&A \cap E^c = f^{-1}(V)
\end{aligned}$$

Since E^c and V are open, by the open-set characterization of continuity, we have f continuous on A.

Now, suppose f continuous on A. We have to show (1).
By the open-set characterization of continuity, we can take arbitrary open $V \subseteq \mathbb{R}^m$ and let open $U \subseteq \mathbb{R}^n$ such that

$$f^{-1}(V) = A \cap U$$

By [a], $f^{-1}(V^c) = A \cap (f^{-1}(V))^c$

$$\begin{aligned}
f^{-1}(V^c) &= A \cap (A \cap U)^c \\
f^{-1}(V^c) &= A \cap (A^c \cup U^c) \\
f^{-1}(V^c) &= (A \cap A^c) \cup (A \cap U^c) \\
f^{-1}(V^c) &= (A \cap U^c)
\end{aligned}$$

Since U and V are open, their complements are closed so we have shown (1) for arbitrary sets. □

[c] Show that $B = \{(x, y) \in \mathbb{R}^2 : xy = 1\}$ is a closed subset of $\mathbb{R}^2$. *Hint:* Consider $g(x, y) = xy$.

Answer:

Lemma. *Let $g : \mathbb{R}^2 \to \mathbb{R}$ such that $g(x, y) = xy$. g is continuous on $\mathbb{R}^2$.*
Let arbitrary $p = (p_1, p_2) \in \mathbb{R}^2$. Take a sequence $(s_k)_{k\in\mathbb{N}} = (x_k, y_k)_{k\in\mathbb{N}}$ that converges to p.
Therefore, $(x_k) \to p_1$ and $(y_k) \to p_2$

$$
\begin{aligned}
g(x_k, y_k) &= x_k y_k \\
\lim_{k\to\infty} g(x_k, y_k) &= \lim_{k\to\infty} x_k y_k \\
\lim_{k\to\infty} g(x_k, y_k) &= p_1 p_2 \\
\lim_{k\to\infty} g(x_k, y_k) &= g(p)
\end{aligned}
$$

So $(g(s_k))$ converges to $g(p)$. Therefore, g is continuous at p. So g is continuous on $\mathbb{R}^2$.

Lemma. *$\{1\} \subseteq \mathbb{R}$ is closed.*
Let arbitrary $x \in \mathbb{R} \setminus \{1\}$. Take $\varepsilon = |x - 1| > 0$. Then, $B_\varepsilon(x) \subseteq \mathbb{R} \setminus \{1\}$.So $\mathbb{R} \setminus \{1\}$ is open. So $\{1\} \subseteq \mathbb{R}$ is closed.

Proof. Let $g : \mathbb{R}^2 \to \mathbb{R}$ such that $g(x, y) = xy$. Since g is continuous and $\{1\}$ is closed in $\mathbb{R}$, part [b] implies that $B = g^{-1}(\{1\})$ is closed in $\mathbb{R}^2$. □

[2] Let $f : A \subseteq \mathbb{R}^n \to \mathbb{R}^m$ be a map. For $\mathbf{x} \in A$, we have

$$f(\mathbf{x}) = \big(f_1(\mathbf{x}), \ldots, f_m(\mathbf{x})\big),$$

where $f_j : A \subseteq \mathbb{R}^n \to \mathbb{R}$ is the jth *component function* of f, for $1 \leq j \leq m$.

[a] Let $\mathbf{x}_0 \in A$. Prove using the sequential characterization of continuity that f is continuous at $\mathbf{x}_0$ if and only if each f_j is continuous at $\mathbf{x}_0$ for all $1 \leq j \leq m$.

Answer:

Proof.

$$\begin{aligned}
&f \text{ is continuous at } x_0 \\
&\iff \forall (x_k)_{k\in\mathbb{N}} \subseteq A : \lim_{k\to\infty} x_k = x_0 \implies \lim_{k\to\infty} f(x_k) = f(x_0) \\
&\iff \forall (x_k)_{k\in\mathbb{N}} \subseteq A : \lim_{k\to\infty} x_k = x_0 \implies \forall j \in [m] : \lim_{k\to\infty} f_j(x_k) = f_j(x_0) \\
&\iff \forall j \in [m] : \forall (x_k)_{k\in\mathbb{N}} \subseteq A : \lim_{k\to\infty} x_k = x_0 \implies \lim_{k\to\infty} f_j(x_k) = f_j(x_0) \\
&\iff \forall j \in [m] : f_j \text{ is continuous at } x_0
\end{aligned}$$

We've shown the step going from line $2 \to 3$ of this in $A2$. □

[b] Let $\pi_j : \mathbb{R}^m \to \mathbb{R}$ be the map $\pi_j(y_1, \ldots, y_m) = y_j$, the projection onto the jth component. We proved in class directly using the definition of continuity that π_j is continuous on $\mathbb{R}^m$. Let $\iota_j : \mathbb{R} \to \mathbb{R}^m$ be the map $\iota_j(t) = (0, \ldots, 0, t, 0, \ldots, 0)$ where the t is in the jth position. Prove directly using the definition of continuity that ι_j is continuous on $\mathbb{R}$.

Answer:

Proof. Let arbitrary $p \in \mathbb{R}$. Let arbitrary $\varepsilon > 0$. Take $\delta = \varepsilon$.
We can see that

$$\|\iota_j(x) - \iota_j(p)\| = \|(0, \ldots, 0, x - p, 0, \ldots, 0)\| = |x - p|$$

Now,

$$\forall x \in \mathbb{R} : \ |x - p| < \delta = \varepsilon \implies \|\iota_j(x) - \iota_j(p)\| = |x - p| < \varepsilon$$

So ι_j is continuous on $\mathbb{R}$ □

[c] Find a way to use the maps π_j, ι_j in [b] to give another proof that f is continuous on A if and only if each f_j is continuous on A for $1 \leq j \leq m$.

Answer:

Proof. We can easily see that

$$\forall j \in [m] : \ (\pi_j \circ f)(x) = f_j(x) \tag{2}$$

And that

$$f(x) = \sum_{j=1}^{m} (\iota_j \circ f_j)(x) \tag{3}$$

We also know from class that

- composition of continuous maps are continuous;
- liniar combinations of continuous maps are continuous;

First, assume f is continuous on A. We also have that π_j is continuous on $\mathbb{R}^m$. Therefore, $(\pi_j \circ f)$ is continuous on A. So, by (2), $\forall j \in [m] :\ f_j$ continuous on A.
Secondly, assume f_j is continuous on A for all $j \in [m]$. ι_j is also continuous on $\mathbb{R}$, so $(\iota_j \circ f_j)$ is continuous on A for all $j \in [m]$.
$\sum_{j=1}^{m}(\iota_j \circ f_j)(x)$ is a liniar combination of continuous maps so it's also continuous. Therefore, by (3), f is continuous on A. $\square$

[3] Let $\mathbb{R}^{n^2}$ denote the space of $n \times n$ (square) matrices. Recall we can multiply two $n \times n$ matrices A, B to obtain a matrix AB.

[a] Let $A, B \in \mathbb{R}^{n^2}$. Recall that the Euclidean inner product on this space is $\langle A, B \rangle = \sum_{i=1}^{n} \sum_{j=1}^{n} A_{ij} B_{ij}$. Prove that $\langle A, B \rangle = \text{tr}(A^T B)$, where A^T is the transpose of A, and $\text{tr}(C)$ is the trace of C.

Answer:

Proof.

$$(A^T B)_{ii} = \sum_{j=1}^{n} A^T_{ij} B_{ji} = \sum_{j=1}^{n} A_{ji} B_{ji}$$

$$tr(A^T B) = \sum_{i=1}^{n} (A^T B)_{ii} = \sum_{i=1}^{n} \sum_{j=1}^{n} A_{ji} B_{ji} = \langle A, B \rangle$$

□

[b] Consider the map $g : \mathbb{R}^{n^2} \to \mathbb{R}^{n^2}$ given by $g(A) = A^T A$. Prove that g is continuous on $\mathbb{R}^{n^2}$.

Answer:

Proof. Let arbitrary $M \in \mathbb{R}^{n^2}$. Let arbitrary sequence $(A_k)_{k \in \mathbb{N}}$ in $\mathbb{R}^{n^2}$ such that

$$\lim_{k \to \infty} A_k = M$$

Now, we've shown previously that for $\mathbb{R}^n$ convergence must happen component-wise, and it is trivial that this is also true in $\mathbb{R}^{n^2}$. So:

$$\lim_{k \to \infty} A_k = M \iff \forall i, j \in [n] : \lim_{k \to \infty} (A_k)_{ij} = M_{ij}$$

Now consider $(g(A_k))_{ij}$:

$$\begin{aligned}
(g(A_k))_{ij} = (A_k^T A_k)_{ij} &= \sum_{r=1}^{n} (A_k)_{ri} (A_k)_{rj} \\
\lim_{k \to \infty} (g(A_k))_{ij} &= \sum_{r=1}^{n} \lim_{k \to \infty} (A_k)_{ri} (A_k)_{rj} \\
&= \sum_{r=1}^{n} M_{ri} M_{rj} \\
&= (g(M))_{ij}
\end{aligned}$$

And this holds $\forall i, j \in [n]$, so

$$\lim_{k \to \infty} g(A_k) = g(M)$$

Therefore, g is continuous at M and since M was arbitrary, g is continuous on $\mathbb{R}^{n^2}$. □

[c] Prove that the subset of $n \times n$ *orthogonal* matrices is a *compact* subset of $\mathbb{R}^{n^2}$.

Answer:

Proof. Let

$$\mathbb{E} = \{A \in \mathbb{R}^{n^2} : \ A^T = A^{-1}\}$$

Equivalently,

$$A \in \mathbb{E} \iff A^T A = I$$

Therefore,

$$\mathbb{E} = g^{-1}(\{I_n\})$$

Since g is continuous and $\{I_n\}$ is a closed singleton set, from $1[b]$ we get that E is also closed.
Now take arbitrary $A \in \mathbb{E}$.

$$\begin{aligned} &A^T A = I_n \\ &tr(A^T A) = tr(I_n) = n \\ &\langle A, A \rangle = n \\ &\|A\| = \sqrt{n} \end{aligned}$$

So $\mathbb{E}$ is bounded by $\sqrt{n}$.
Therefore, $\mathbb{E}$ is closed and bounded, so it is compact. □

[4] Let $A \subseteq \mathbb{R}^n$ be nonempty, and fix a point $\mathbf{x} \in \mathbb{R}^n$.

[a] If A is *compact*, show that there exists at least one point $\mathbf{y} \in A$ such that $\|\mathbf{y} - \mathbf{x}\| \geq \|\mathbf{a} - \mathbf{x}\|$ for all $\mathbf{a} \in A$. That is, $\mathbf{y}$ is a point in A that is *furthest* from $\mathbf{x}$.
Hint: Find a way to use the Extreme Value Theorem.

Answer:
Consider the map $f : A \subseteq \mathbb{R}^n \to \mathbb{R}$ given by

$$f(a) = \|a - x\| = \sqrt{\sum_{i=1}^{n} (a_i - x_i)^2}$$

We want to show that f is continuous.
Let arbitrary $a \in A$, let arbitrary $\varepsilon > 0$. Choose $\delta = \varepsilon$.
Take arbitrary $b \in A$ such that $\|b - a\| < \delta$.
From the triangle inequality, we get that

$$|\|b - x\| - \|a - x\|| \leq \|b - a\|$$

So

$$|\|b - x\| - \|a - x\|| \leq \|b - a\| < \delta = \varepsilon$$
$$|\|b - x\| - \|a - x\|| < \varepsilon$$
$$|f(b) - f(a)| < \varepsilon$$

So f is continuous on A and A is compact. By the Extreme Value Theorem,

$$\exists y \in A : \ \forall a \in A : \ f(y) \geq f(a)$$

So

$$\exists y \in A : \ \forall a \in A : \ \|y - x\| \geq \|a - x\|$$

[b] Give an example to show that $\mathbf{y}$ is not unique in general. (There are many possible answers.)

Answer:
Let $A = [-1, 1]$ and $x = 0$. Then y can be both -1 and 1 as both achieve the max distance from x.
Also we should note that A is compact because it is a closed interval in $\mathbb{R}$.

[c] If A is *not compact*, there may not exist any such $\mathbf{y}$. If A is noncompact because it is unbounded, then this is trivial to see, as there are points in A arbitrarily far from the origin, and hence arbitrarily far from any $\mathbf{x}$. More interesting is if A is noncompact because it is bounded but *not closed*. Give an example of such an A and $\mathbf{x}$ such that there is no farthest point in A from $\mathbf{x}$. (There are many possible answers.)

Answer:
Let $A = (0, 1)$ and $x = 0$. In this case, there is no y that achieves the maximum distance away from x as you can always get closer to 1.

[5] Let $f : \mathbb{R}^n \to \mathbb{R}$ be continuous on $\mathbb{R}^n$. Suppose that f has the following property: for any $\varepsilon > 0$, there exists $M > 0$ such that if $\|\mathbf{x}\| > M$, then $|f(\mathbf{x})| < \varepsilon$. (We say that f *goes to zero at infinity.*)

[a] Show that f is bounded on $\mathbb{R}^n$.

Answer:

Proof. Take $\varepsilon = 1$. There exists $M > 0$ such that

$$\|x\| > M \implies |f(x)| < 1$$

Now consider the closed ball $\overline{B_M(0)}$.

$$x \in \overline{B_M(0)} \iff \|x\| \leq M$$

f is continuous and $\overline{B_M(0)}$ is closed and bounded, so $f(\overline{B_M(0)})$ is also bounded.
Let K be the bound of $f(\overline{B_M(0)})$.
Now, if $\|x\| \leq M$, then $f(x)$ bounded by K and if $\|x\| > M$, $f(x)$ bounded by $\varepsilon = 1$.
Therefore, f bounded by $\max K, 1$. □

[b] Suppose there exists some point $\mathbf{x}_0 \in \mathbb{R}^n$ such that $f(\mathbf{x}_0) > 0$. Prove that there exists a point $\tilde{\mathbf{x}}$ such that $f(\mathbf{x}) \leq f(\tilde{\mathbf{x}})$ for all $\mathbf{x} \in \mathbb{R}^n$. (That is, show that f *attains a global maximum* on $\mathbb{R}^n$.)
[Note that we cannot *directly* use the Extreme Value Theorem because $\mathbb{R}^n$ is not compact. Find a creative way to use it nevertheless.]

Answer:

Proof. Take $\varepsilon = f(x_0) > 0$. There exists $M > 0$ such that

$$\|x\| > M \implies |f(x)| < f(x_0)$$

Therefore, for any x ouside the closed ball of size M centered at 0, $f(x)$ is bounded by $f(x_0)$.
For all x inside the closed ball, we can apply the EVT on that subset to show that there exists $\bar{x}$ for which $f(x) \leq f(\bar{x})$
Now simply take $\tilde{x} \in \{x_0, \bar{x}\}$ for which $f(\tilde{x}) = \max\{f(x_0), f(\bar{x})\}$. □

[c] Show that the assumption that there exists $\mathbf{x}_0 \in \mathbb{R}^n$ with $f(\mathbf{x}_0) > 0$ is necessary for part [b]. That is, find an example where f *goes to zero at infinity*, with $f(\mathbf{x}) \leq 0$ for all $\mathbf{x} \in \mathbb{R}^n$, and such that f does not attain a global maximum on $\mathbb{R}^n$. [Take $n = 1$ for simplicity.]

Answer:
Let $f(x) = -e^{-x^2}$. $f(x) \leq 0$ for all x and $\lim_{x\to\infty} f(x) = 0$.
f does not have a global maximum.

Proof. Assume for the sake of contradiction that f has a global maximum. Then,

$$\exists \tilde{x} : \ \forall x \in \mathbb{R} : \ f(\tilde{x}) \geq f(x)$$

However

$$\begin{aligned}
&\tilde{x}+1>\tilde{x}\\
&(\tilde{x}+1)^2>\tilde{x}^2\\
&-(\tilde{x}+1)^2<-(\tilde{x})^2\\
&e^{-(\tilde{x}+1)^2}<e^{-(\tilde{x})^2}\\
&-e^{-(\tilde{x}+1)^2}>-e^{-(\tilde{x})^2}\\
&f(\tilde{x}+1)>f(\tilde{x})
\end{aligned}$$

Which is a contradiction. □